

Phase 3 - Batch_3 baseline

Batch_3 baseline (17 images): visible pore fraction 0.114 (5-95% 0.093-0.154), silicon 0.067, graphite 0.819. Internal consistency: 7 image(s) sit >3.5 robust-z on some feature - the batch is broadly homogeneous with a high-contrast subgroup; no image was removed from the baseline for that reason alone.

Assumption (applies throughout): the grey bulk is assumed graphite, bright particles are assumed silicon-based, black is pore/crack. This comes from image appearance and is unconfirmed.

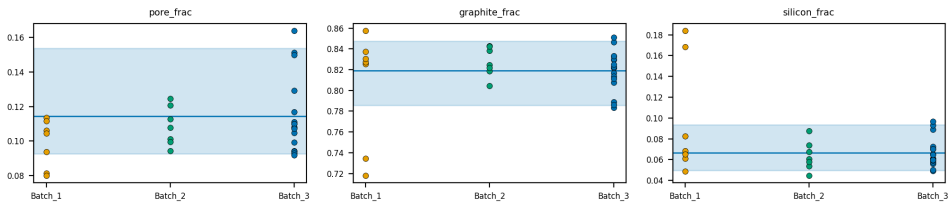
feature	n	mean	median	std	p5	p95	ci95_lo	ci95_hi
pore_frac	17	0.1142	0.108	0.02174	0.09273	0.1537	0.1054	0.1248
graphite_frac	17	0.819	0.8218	0.01923	0.7856	0.8473	0.8098	0.8275
silicon_frac	17	0.06676	0.0611	0.01404	0.04988	0.09367	0.06094	0.07313
si_d10_um	17	0.385	0.377	0.024	0.3617	0.4241	0.3749	0.396
si_d50_um	17	0.763	0.6852	0.2052	0.5653	1.123	0.6763	0.8627
si_solidity	17	0.8959	0.8981	0.01772	0.8635	0.9128	0.8876	0.903
gr_chord_h_um	17	3.668	3.717	0.4279	2.966	4.236	3.471	3.856
gr_chord_ratio_hv	17	1.2	1.192	0.03784	1.151	1.263	1.183	1.217
gr_st_coherence	17	0.1084	0.11	0.008118	0.09391	0.1181	0.1045	0.1118
pore_thick_d10_um	17	0.05365	0.05	0.008139	0.05	0.07071	0.05	0.05731
pore_thick_d50_um	17	0.2697	0.25	0.04598	0.2213	0.3674	0.2511	0.2921
large_gap_frac	17	0.01476	0.008498	0.02019	0	0.0599	0.006908	0.02473
pores_per_mm2	17	1.55e+05	1.55e+05	1.298e+04	1.371e+05	1.726e+05	1.489e+05	1.607e+05
ring_porosity_250nm	17	0.08355	0.08289	0.02158	0.05942	0.1163	0.07462	0.0939
ring_porosity_500nm	17	0.2	0.189	0.04069	0.1583	0.2659	0.1837	0.2188
corr_len_gr_um	17	0.9353	0.95	0.1042	0.75	1.06	0.8882	0.9824
si_clarkevans_R	17	0.6793	0.6758	0.04339	0.6133	0.7364	0.6599	0.6978
inlens_edge_density	17	0.04015	0.04204	0.007468	0.02971	0.04955	0.03666	0.04343
inlens_bulk_texture	17	50.93	50.91	11.25	38.21	64.91	45.65	56.34
etd_roughness	17	0.0303	0.02973	0.003461	0.02595	0.03465	0.02872	0.03187
bse_bulk_texture	17	11.82	12.78	1.728	9.66	13.7	11	12.6

Internal consistency (robust z per image)

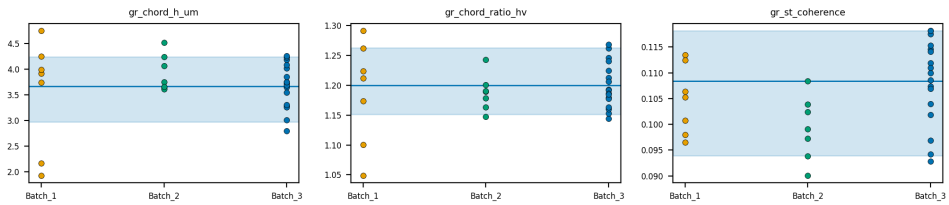
image_id	max_abs_robust_z	mean_abs_robust_z	outlier
0grcilhi	2.372e+04	1131	True
71vgq3fw	2.222	0.6781	False
9luzk4jm	2.372e+04	1131	True
cfe5vt7s	1.22	0.4721	False
hawkfj64	2.184	0.6731	False
hzumfsms	2.372e+04	1131	True

image_id	max_abs_robust_z	mean_abs_robust_z	outlier
kbdh4tri	3.645	1.186	True
mgxahqnk	2.019	0.6263	False
pl8uabbv	1.043	0.419	False
ptg8lmt0	1.922	0.6864	False
tuy3zymq	4.243	1.143	True
ufdvpb81	2.346	0.7655	False
utfgcjfa	2.16	0.7066	False
vc2whyay	3.212	0.968	False
x77cy643	3.766	0.8191	True
x7u69zsw	4.64	0.9597	True
xgj4xftb	1.348	0.5581	False

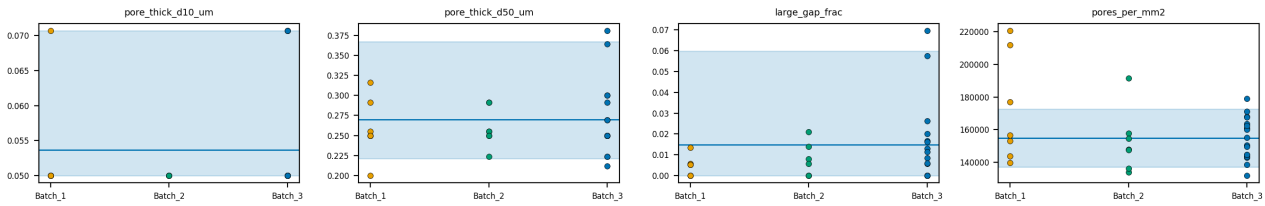
composition features - points vs Batch_3 5-95% band



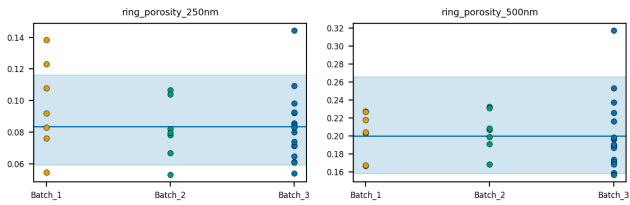
graphite features - points vs Batch_3 5-95% band



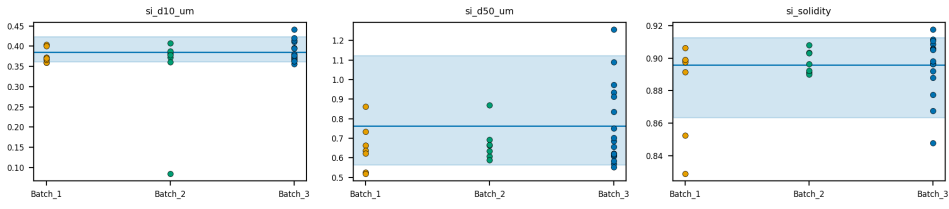
pores features - points vs Batch_3 5-95% band



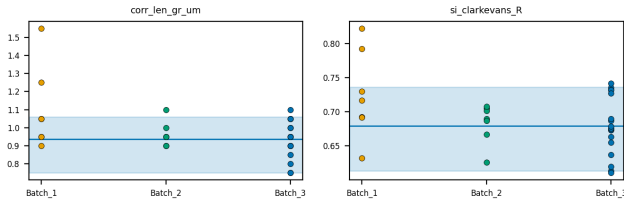
si_pore features - points vs Batch_3 5-95% band



silicon features - points vs Batch_3 5-95% band



spatial features - points vs Batch_3 5-95% band



texture features - points vs Batch_3 5-95% band

